SAFAR KAABE

COMPLETE TRAVEL BOOKING MARKETPLACE

One platform. All travel agencies. Every journey.

Detailed Product Requirements & Technical Implementation Plan

Web Application • Android & iOS • Windows & macOS Desktop

Business model: Online marketplace connecting travelers, travel agencies, airlines, hotels, car rental companies, tour operators, and transfer providers.

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Planning document. External supplier and payment integrations require provider approval, valid credentials, and testing before production use.

Document Contents

- 1. Executive Summary and Business Model
- 2. Complete Platform Scope
- 3. Recommended Technical Architecture
- 4. Repository and Application Structure
- 5. Customer Website and Mobile Screens
- 6. Agency Portal
- 7. Supplier Portal
- 8. Central Administration Dashboard
- 9. Desktop Application
- 10. User Roles and Permissions
- 11. Database Design and Data Relationships
- 12. REST API Endpoint Specification
- 13. External Integrations and Supplier Strategy
- 14. Booking Engine and Financial Logic
- 15. Security, Privacy, and Compliance
- 16. Deployment and Infrastructure
- 17. Testing and Acceptance Criteria
- 18. Phased Development Roadmap
- 19. Recommended Team
- 20. Risks and Mitigations
- 21. Project Kickoff Checklist
- 22. Final Delivery and Acceptance Checklist

This document consolidates the SAFAR KAABE requirements and implementation plan into a developer-oriented specification. It includes the full platform scope, workflows, technical recommendations, data model, API contract outline, integration requirements, security controls, release plan, and acceptance criteria.

1. Executive Summary and Business Model

SAFAR KAABE is a proposed multi-agency online travel booking marketplace. It brings travelers and travel agencies together with airlines, hotels, car rental companies, tour operators, transfer providers, and package operators through one digital platform.

The intended product includes a customer-facing website, Android and iOS apps, a dedicated desktop application for Windows and macOS, a central administration dashboard, a travel-agency portal, and a supplier portal. All applications use a shared backend and authoritative booking database.

Business model

- **Booking commissions:** Commission earned on eligible completed bookings.
- **Agency subscriptions:** Basic, Professional, and Enterprise plans for participating agencies.
- **Service fees:** Clearly disclosed booking or processing fees where permitted.
- **Supplier agreements:** Commercial rates and commissions negotiated with travel providers.
- **Featured listings:** Clearly labeled sponsored placements for eligible agencies or suppliers.
- **Corporate travel:** Managed bookings and invoicing for businesses and organizations.

The financial model must distinguish supplier price, agency commission, SAFAR KAABE fee, payment-processing costs, taxes, refunds, settlement amounts, and recognized revenue. Commission, markup, and resale rights vary by supplier contract.

Platform principles

- One shared backend and data model across the supported applications.
- Multi-agency and multi-supplier architecture with strict tenant isolation.
- Accurate booking states and honest distinction between live offers, pending requests, and confirmed reservations.
- Modular architecture that is simple to launch and can scale over time.
- No fake payment success, fabricated availability, or simulated production confirmations.

2. Complete Platform Scope

Customer and travel features

- Registration and sign-in by email or phone, verification, password reset, optional multi-factor authentication, profiles, preferences, and saved travelers.
- Flight search for one-way, round-trip, and multi-city journeys; fare details, baggage, fare rules, and ticket information.
- Hotel search with dates, guests, rooms, amenities, rates, availability, policies, and property details.
- Car rental search with pickup and return locations, dates, vehicle classes, driver requirements, and deposits.
- Tours and activities with schedules, capacity, participant details, itineraries, vouchers, and cancellation terms.
- Transfers with pickup/drop-off locations, flight arrival details where relevant, luggage, vehicle class, and service confirmation.
- Travel packages combining eligible services with component-level availability, pricing, and cancellation rules.
- Search filters, sorting, maps, reviews, favorites, destination pages, and offer comparisons.
- Checkout, price revalidation, booking references, payment, confirmations, tickets, vouchers, invoices, and trip history.
- Changes, cancellation requests, refund tracking, customer support, loyalty programs, coupons, and price alerts where supported.
- Somali and English at launch, with architecture that supports further languages.
- Multi-currency display with explicit currency, exchange-rate basis, and transparent fees.
- Downloadable tickets and itineraries with offline access to previously saved documents.

Business and administration features

- Agency registration, verification, staff accounts, permissions, booking workspace, customer CRM, commissions, statements, and subscriptions.
- Supplier onboarding, contracts, eligible inventory, rate plans, schedules, capacity, booking inbox, settlements, and integration status.
- Central dashboard for bookings, customers, suppliers, agencies, payments, refunds, commissions, content, and support.
- Content management for destinations, listings, FAQs, banners, blogs, and SEO fields.
- Customer support tickets, live chat or ticketing integration, complaint escalation, and resolution history.
- Reporting for gross booking value, revenue, refunds, commissions, supplier costs, conversion, and engagement.
- Audit trails, access controls, integration monitoring, automated backups, and operational alerts.

Advanced features

- AI-based travel recommendations where sufficient data and appropriate consent exist.
- Price alerts and fare tracking where supplier terms and API capabilities permit.
- Travel insurance integration through suitable authorized partners.
- Visa and entry information from reliable sources, with clear disclaimers and update dates.

- Weather, maps, local guides, accessibility information, and group/family booking.
- Sustainability filters and virtual tours when reliable data or supported providers are available.

3. Recommended Technical Architecture

Start with a modular monolith: one well-structured backend application divided into clear business modules. This reduces initial complexity while leaving the option to extract high-load or independently operated modules into services later.

Layer	Recommended technology	Responsibility
Web marketplace and admin	Next.js, React, TypeScript	Customer discovery, checkout, agency/supplier/admin portals
Mobile apps	React Native + Expo	Android and iOS customer and operational experiences
Desktop app	Electron	Installable Windows/macOS application with secure local storage and updates
Backend	NestJS + TypeScript	Business rules, APIs, authorization, integrations, booking orchestration
API contract	REST + OpenAPI	Versioned, documented API and generated clients
Database	PostgreSQL	Authoritative records for users, organizations, bookings, finance, audit
Cache and jobs	Redis + BullMQ	Caching, asynchronous work, retries, scheduled jobs
File storage	S3-compatible object storage	Tickets, vouchers, invoices, images, private documents
Deployment	Docker, managed cloud, CI/CD	Reproducible builds, controlled releases, monitoring

Logical system flow

Customer website / mobile / desktop → versioned backend API → business modules → PostgreSQL and Redis/job workers → supplier and payment adapters → notifications and monitoring.

Architecture requirements

- Use shared API contracts and typed validation across clients.
- Keep supplier credentials and payment secrets on the server, never in app binaries.
- Use provider adapters behind normalized internal interfaces.
- Use asynchronous jobs for notifications, supplier polling, reconciliation, and retryable tasks.
- Make booking and payment operations idempotent where possible.
- Maintain separate development, staging, and production environments.
- Document architecture decisions and integration capabilities.

4. Repository and Application Structure

```
safar-kaabe/
apps/
web/ # Customer marketplace
admin/ # Central administration
agency-portal/ # Agency workspace
supplier-portal/ # Supplier workspace
mobile/ # Android and iOS
desktop/ # Windows and macOS
api/ # Backend REST API
packages/
ui/ # Shared design system
types/ # Shared TypeScript types
validation/ # Shared input validation
api-client/ # Generated API client
config/ # Shared configuration
booking-domain/ # Common booking rules
infrastructure/
docker/
deployment/
monitoring/
database-backups/
docs/
requirements/
api/
architecture/
integrations/
security/
user-guides/
tests/
integration/
end-to-end/
performance/
.env.example
package.json
pnpm-workspace.yaml
README.md
```

A workspace manager such as pnpm with Turborepo or an equivalent tool can coordinate builds. Do not commit real secrets, API keys, signing keys, database passwords, or production credentials to the repository.

Suggested backend modules

- Identity and access management
- Organizations, agency tenancy, supplier onboarding, and verification
- Destination catalog and product normalization
- Search orchestration and provider adapters
- Quotes, booking orchestration, supplier reservations, and booking history
- Payments, refunds, commissions, invoices, settlements, and reconciliation
- Notifications and scheduled jobs
- Reviews, loyalty, promotions, content management, and support
- Administration, audit logging, analytics, and reporting

5. Customer Website and Mobile Screens

Discovery and search

- Home page with search tabs, destination content, featured offers, and clear booking entry points.
- Flight search: origin, destination, dates, traveler count, cabin, and one-way/round-trip/multi-city.
- Hotel search: destination, check-in/check-out, guests, rooms, price, rating, and amenities.
- Car search: pickup/return location and time, vehicle class, driver eligibility, and rental conditions.

- Tours and activities: destination, date, category, participant count, duration, and language.
- Transfers: pickup/drop-off, date/time, passenger and luggage count, and vehicle type.
- Packages: destination, travel dates, travelers, included services, and customization where supported.
- Results screens with filters, sorting, map view, comparison, loading states, no-result states, and supplier availability warnings.

Product details and checkout

- Flight itinerary, fare rules, baggage allowance, connection details, traveler requirements, and supplier terms.
- Hotel photos, room selection, occupancy, taxes, amenities, accessibility, and cancellation policies.
- Car specifications, insurance/deposit terms, mileage, pickup instructions, and driver requirements.
- Tour itinerary, inclusions/exclusions, meeting point, schedule, capacity, and voucher rules.
- Transfer vehicle, pickup instructions, waiting-time policy, and flight tracking where supported.
- Package components, combined price, availability dependencies, and component cancellation terms.
- Traveler details, contact information, optional extras, coupon codes, final price breakdown, payment method, and terms acceptance.
- Booking success, pending status, failure recovery, confirmation reference, ticket/voucher, invoice, and notifications.

Account and support

- Registration, login, OTP verification, password reset, optional MFA, and active session management.
- Profile, locale, currency, saved travelers, preferences, and privacy controls.
- My trips, booking details, documents, cancellation requests, changes, refunds, and booking history.
- Favorites, reviews for eligible completed bookings, loyalty points, and price alerts.
- Help center, FAQs, support tickets, contact page, notifications, and account deletion request.
- Privacy policy, terms, refund rules, cookie consent where applicable, accessibility, and destination guides.

6. Agency Portal

- Agency application, identity/business verification, documents, profile, and approved branding settings.
- Agency owner and staff invitations, role assignment, session management, and access logs.
- Unified search and eligible booking tools for customers served by the agency.
- Customer CRM and traveler profiles with access restrictions and appropriate consent.
- Create and manage permitted reservations; view supplier references, status, and documents.
- Handle changes, cancellations, refund requests, customer communications, and supplier escalations.
- Commission, markup, subscription, invoices, statements, payout information, and reconciliation views.
- Sales performance and operational reports.
- Support tickets, integration limitations, billing, and account settings.

Each agency needs a tenant boundary. An agency must not see another agency's private customers, bookings, negotiated rates, or financial information unless an explicitly authorized shared workflow exists.

7. Supplier Portal

- Supplier registration, verification, profile, contracts, service categories, and capabilities.
- Staff access and permission management.
- Product/listing management for supplier-managed inventory.
- Rates, availability calendars, capacity, blackout dates, restrictions, and cancellation rules.
- Tour departures, vehicle fleets, transfer routes, pickup locations, and package components where applicable.

- Booking inbox, confirmations, operational updates, and permitted cancellation/change actions.
- Supplier invoices, payout/settlement records, reconciliation, and performance reports.
- API connection status, error diagnostics, and webhook status where an API integration exists.

Airlines and other API-only suppliers may not support manual inventory management in this portal. Supplier features must reflect each provider's contract and actual technical capabilities.

8. Central Administration Dashboard

Area	Required capabilities
Overview	Gross booking value, recognized revenue, commissions, bookings, refunds, conversion, supplier health
Marketplace	Agencies, suppliers, verification, products, destinations, categories, featured listings
Booking operations	Search bookings, inspect supplier references, manage exceptions, track state transitions
Finance	Payments, refunds, disputes, commissions, invoices, supplier settlements, reconciliation
Support	Tickets, messages, escalation, service-level targets, resolution history
Marketing/CMS	Banners, destinations, blogs, FAQs, SEO metadata, coupons, campaigns
Security	Roles, permissions, audit logs, privileged actions, integration configuration
Operations	Background job health, supplier errors, failed webhooks, alerts, feature flags
Reporting	Date filters, export permissions, agency and supplier reports, financial summaries

Sensitive operations such as refund approvals, payout changes, role elevation, and supplier credential updates must be restricted and audited. Consider dual approval for high-value refunds and payouts.

9. Dedicated Desktop Application

- Secure login and authorized agency workspace selection.
- Unified travel search and comparison.
- Customer CRM and traveler profiles.
- Booking management, reservation history, supplier references, and status tracking.
- Payment status, invoice viewing, ticket and voucher document management.
- Support inbox and operational notifications.
- Offline access to previously downloaded itineraries and documents.
- Application settings, signed updates, secure logout, and version information.

Offline mode must never claim a live booking, payment, cancellation, or refund succeeded. Actions requiring server confirmation must remain clearly pending until confirmed.

Electron is the recommended cross-platform desktop option. Reuse shared components when practical, but test installation, signing, updates, secure local storage, and platform behavior separately on Windows and macOS.

10. User Roles and Permissions

Role	Main permissions
Traveler	Search, book, pay, view own trips, submit support requests
Agency owner	Manage agency profile, staff, bookings, permitted financial reports
Agency manager	Manage authorized agency operations and bookings
Agency agent	Search, create permitted bookings, support assigned customers
Supplier owner	Manage supplier account, eligible inventory, and reports
Supplier staff	Manage assigned services and supplier bookings
Customer support	View authorized booking details, handle cases, escalate issues
Finance officer	Reconcile payments, refunds, commissions, and settlements
Operations manager	Handle booking exceptions, supplier issues, and operational queues
Content manager	Manage destination content, listings, banners, and promotions
Super administrator	Manage authorized platform configuration and privileged operations
Auditor	Read permitted audit/financial records without changing them

Authorization requirements

- Use role-based access control plus organization/tenant-level resource restrictions.
- Enforce authorization on the backend for every protected operation; hiding a button is not security.
- Restrict supplier users to their own records and bookings they are authorized to fulfill.
- Restrict agency users to their organization's records and permitted customers.
- Use stronger authentication and audit logs for refunds, payouts, privilege changes, and credential management.

11. Database Design and Data Relationships

Entity/table	Purpose
users; user_profiles	Login identities, contact details, preferences, locale
roles; permissions; user_roles	Authorization model
organizations; organization_members	Agency, supplier, corporate organization and membership
agency_profiles; supplier_profiles	Business profile, verification, commercial settings
supplier_connections	Integration configuration and status; secrets referenced from secure vault
destinations	Countries, cities, airports, destination content
travel_products	Normalized product catalog and supplier ownership
flight_offers	Flight offer references, expiry, fare and price snapshots
hotel_properties; hotel_rooms; hotel_rates	Accommodation catalog, rooms, rates
vehicles; rental_locations	Car rental inventory and locations
tours; tour_departures	Activities, schedules, capacity
transfer_routes; transfer_options	Pickup/drop-off and transfer offers
packages; package_components	Packages and linked travel components
search_sessions	Search criteria and provider response references
bookings; booking_items	Reservations and purchased services
travelers; booking_travelers	Traveler profiles and booking associations
supplier_reservations	Provider references and confirmation states
booking_status_history	Traceable booking state transitions
payments; payment_attempts	Payment state and provider references
refunds; chargebacks	Refund lifecycle and payment disputes
commissions; fees; settlements	Agency/platform financial calculations and settlement records
invoices; invoice_items	Customer and supplier billing
coupons; promotions; loyalty_accounts	Offers, discounts, rewards
reviews; favorites	Customer feedback and saved products
notifications; notification_preferences	Delivery history and preferences
support_tickets; support_messages	Support workflow and communications
audit_logs	Sensitive actions and changes
webhook_events; integration_jobs	Event processing, retries, diagnostics

Relationship model

Organization (agency or supplier) → Products/offers → Booking → Booking items → Supplier reservations. A booking can also relate to payments, refunds, commissions, settlements, invoices, travelers, documents, notifications, and immutable status history.

Database implementation rules

- Use stable UUIDs or equivalent identifiers for entities.

- Store money as fixed-precision decimals or integer minor units with explicit ISO currency codes.
- Preserve supplier price, customer-facing price, markup, fee, tax, commission calculation, and currency for each booking.
- Store timestamps in UTC and retain the travel location's time zone.
- Keep historical snapshots of booked product details, prices, and cancellation terms.
- Use appropriate indexes, constraints, transactions, and query-performance testing.
- Apply retention rules to financial/audit records and soft-delete only where appropriate.
- Encrypt sensitive personal and travel-document data and restrict access.
- Store files in object storage and metadata/permissions in PostgreSQL.
- Keep payment and booking status changes traceable and consistent.
- For flights, store supplier offer reference, expiry, fare conditions, and price snapshot; revalidate before purchase.

12. REST API Endpoint Specification

Proposed API base path: **/api/v1**. All endpoints require input validation, consistent errors, appropriate authorization, and pagination for list results. These routes define the intended contract; they are not evidence that the API has already been implemented.

Authentication and customer profiles

Method	Endpoint	Purpose
POST	/auth/register	Create account
POST	/auth/login	Sign in
POST	/auth/logout	End session
POST	/auth/verify-otp	Verify one-time code
POST	/auth/forgot-password	Request password reset
POST	/auth/reset-password	Reset password
GET	/users/me	Get current user profile
PATCH	/users/me	Update permitted profile fields
GET / POST	/users/me/travelers	List or add saved travelers
PATCH / DELETE	/users/me/travelers/{id}	Update/remove saved traveler

Search and catalog

Method	Endpoint	Purpose
GET	/destinations	Browse destinations
GET	/airports	Airport and city lookup
POST	/search/flights	Search flights
POST	/search/hotels	Search hotels
POST	/search/cars	Search rental vehicles
POST	/search/tours	Search tours and activities
POST	/search/transfers	Search transfers
POST	/search/packages	Search packages
GET	/offers/{id}	Retrieve offer details
POST	/offers/{id}/revalidate	Recheck price and availability
GET	/products/{id}	Retrieve permitted product details
GET	/agencies	List eligible public marketplace agencies

Bookings

Method	Endpoint	Purpose
POST	/bookings/quotes	Validate itinerary and calculate price
POST	/bookings	Create booking or booking request
GET	/bookings	List authorized bookings
GET	/bookings/{id}	Booking details
POST	/bookings/{id}/confirm	Advance an eligible booking workflow
POST	/bookings/{id}/cancel	Request cancellation
POST	/bookings/{id}/change-requests	Request change
GET	/bookings/{id}/documents	Retrieve authorized document metadata
GET	/bookings/{id}/status	Check current booking status
GET	/bookings/{id}/history	View status history

Payments and refunds

Method	Endpoint	Purpose
POST	/payments/intents	Start eligible payment
GET	/payments/{id}	Retrieve payment status
POST	/payments/{id}/verify	Verify with provider
POST	/bookings/{id}/refund-requests	Submit refund request
GET	/refunds/{id}	View refund status
POST	/webhooks/payments/{provider}	Receive authenticated payment events
GET	/invoices/{id}	Retrieve authorized invoice

Agency and supplier endpoints

Method	Endpoint	Purpose
POST	/agencies/applications	Apply to join marketplace
GET / PATCH	/agencies/me	Read/update permitted agency information
GET / POST	/agencies/me/members	Manage agency staff
GET	/agencies/me/bookings	View authorized agency bookings
GET	/agencies/me/commissions	View agency commissions
GET	/suppliers/me	Retrieve supplier profile
GET / POST	/suppliers/me/products	Manage eligible products
GET / POST	/suppliers/me/inventory	Manage supported inventory
GET	/suppliers/me/bookings	View supplier's assigned reservations
GET	/suppliers/me/settlements	View permitted settlement records

Administration, support, and reporting

- GET /admin/dashboard
- GET /admin/users
- GET /admin/organizations
- GET /admin/approvals
- GET /admin/bookings
- GET /admin/payments
- GET /admin/refunds
- GET /admin/settlements
- GET /admin/integrations
- GET /admin/audit-logs
- GET /admin/reports/revenue
- GET /admin/reports/bookings
- GET /support/tickets
- GET /support/tickets/{id}/messages
- GET /notifications
- PATCH /notifications/preferences
- POST /webhooks/suppliers/{provider}

Use the correct HTTP method and resource permissions for every route. Do not expose administrative or finance endpoints to ordinary users. Keep passenger details out of URLs and sensitive logs.

13. External Integrations and Supplier Strategy

Real travel inventory and confirmed reservations require authorized supplier APIs, approved partner access, or direct provider agreements. Until credentials and permissions are configured and verified, use clearly labeled mock adapters and sandbox environments.

Service	Integration options	Required capability checks
Flights	Authorized GDS, NDC provider, airline API, flight consolidator	Coverage, search, fare revalidation, booking, ticketing, changes, cancellation, refund
Hotels	Approved hotel inventory API or direct accommodation partners	Availability, room/rate plans, taxes, cancellation, confirmation, refunds
Cars	Authorized rental inventory provider or local rental partners	Locations, driver requirements, availability, deposits, cancellation
Tours	Activity provider API or onboarded local tour operators	Schedules, capacity, participant data, vouchers, cancellation
Transfers	Transfer API or approved local operators	Pickup/drop-off, flight details, luggage, vehicle, confirmation
Packages	Internal orchestration of eligible components	Component availability, price, partial confirmation, cancellation dependency
Payments	Provider approved for the company's market and merchant model	Collection, refunds, settlements, webhooks, currencies, fees, sandbox
Communications	Email, SMS, push, optional messaging integration	Delivery, retries, consent, templates, delivery status
Supporting data	Maps, geocoding, exchange rates, analytics, monitoring	Coverage, licensing, rate limits, accuracy, privacy

Integration implementation sequence

- Research provider coverage, commercial terms, supported countries, and API capabilities.
- Apply for partner access and complete required business verification.
- Obtain sandbox credentials and technical documentation.
- Build a provider adapter behind a common internal interface.
- Map provider responses into the normalized offer model.
- Implement supported search, revalidation, booking, cancellation, and refund operations.
- Test timeouts, duplicates, price changes, provider errors, and webhook processing.
- Complete certification/approval if required.
- Configure production credentials securely and perform controlled live tests.

Payment provider evaluation

- International cards and supported digital wallets where available.
- Somali mobile-money services, including EVC Plus and ZAAD, subject to merchant access and verified technical availability.
- Business registration and settlement-country eligibility.
- Currencies, conversion, fees, refund processing, and settlement timing.
- Webhook signing, sandbox, production approval, and reconciliation support.
- Cross-border payment restrictions and applicable compliance obligations.

Never assume one provider supports all countries, travel services, or booking operations. Maintain an integration capability registry for each supplier and payment provider.

14. Booking Engine and Financial Logic

Recommended booking lifecycle

Stage	Required behavior
1. Search	Retrieve current offers and retain supplier references and expiry.
2. Revalidate	Recheck availability, fare, taxes, and conditions.
3. Quote	Calculate complete customer price and quote expiry.
4. Reserve	Create supplier reservation or supported hold according to provider rules.
5. Pay	Collect and verify payment securely according to the required booking flow.
6. Confirm	Complete supplier booking and required ticketing/issuance.
7. Deliver	Store authorized documents and send customer notifications.

The exact order varies by supplier. Some providers require payment before confirmation; others create a reservation before payment. Flight ticketing can be a separate operation from reservation creation.

Failure scenarios and recovery

Scenario	Required behavior
Price changes	Show updated price and request customer approval before proceeding.
Supplier unavailable	Mark offer unavailable and suggest alternatives where possible.
Payment timeout	Verify provider status before retrying to avoid duplicate collection.
Payment succeeds, booking fails	Run safe recovery or initiate the appropriate refund workflow.
Booking succeeds, response is lost	Query supplier status before attempting another reservation.
Duplicate webhook	Process idempotently; do not duplicate financial entries.
Partial cancellation	Show each component's state and remaining actions.
Refund pending	Show pending status until provider confirmation.

Financial records

- Supplier base price, taxes, and mandatory charges.
- Supplier commission or discount.
- Agency markup/commission and SAFAR KAABE service fee.
- Payment-processing fee, currency conversion, and applicable tax.
- Customer total and amount collected.
- Amount payable to supplier and agency.
- Refund amount, cancellation penalties, chargebacks, and outstanding liabilities.
- Recognized revenue, settlement state, and reconciliation differences.

Accounting warning: Gross booking value is not automatically SAFAR KAABE revenue. The accounting treatment depends on whether the business acts as an agent, reseller, or merchant of record. Agree the model with a qualified accountant.

15. Security, Privacy, and Compliance

- Authentication: secure password hashing, verified email/phone flows, optional MFA, session expiry, and revocation.
- Authorization: role-based access plus organization-level resource restrictions on every protected backend operation.

- Payment security: hosted payment pages or provider tokenization; never store raw card security codes.
- Supplier credentials: managed secret vault or protected configuration with restricted access and rotation.
- Personal data: encryption, least-privilege access, redaction in logs, and retention/deletion rules.
- API protection: validation, rate limiting, abuse detection, safe CORS configuration, and common web vulnerability defenses.
- Webhook protection: signature verification, replay protection, idempotent processing, and event logging.
- Auditability: record sensitive changes, refunds, permissions, payouts, and administrative actions.
- Desktop security: signed releases, verified updates, secure local token storage, and no embedded backend secrets.
- Recovery: encrypted backups, documented restoration procedures, and regular disaster-recovery tests.
- Compliance: assess travel licensing, consumer protection, refunds, tax, privacy, and cross-border payment rules in target markets.

Obtain qualified legal and security review before launch. This plan is not a legal opinion or certification of compliance.

16. Deployment and Infrastructure

Environment	Purpose
Local development	Sample inventory, mock supplier adapters, payment sandbox, fast automated tests
Staging	Production-like configuration, sandbox integrations, staff acceptance testing
Production	Approved supplier credentials, live payments, monitoring, backups, controlled releases

Deployment checklist

- Register brand domain, configure DNS, HTTPS, and appropriate security headers.
- Configure managed application hosting and PostgreSQL.
- Configure Redis, job workers, scheduled tasks, and retry policies.
- Configure object storage and signed URLs for private documents.
- Configure secrets management and separate environment access policies.
- Set up database migrations, backups, and restore tests.
- Configure monitoring, alerts, structured logs, and error tracking.
- Set up automated builds, tests, staging deployments, and production approval.
- Publish mobile apps through relevant app stores.
- Package and sign Windows/macOS desktop releases.
- Document release notes, versioning, rollback, and maintenance ownership.

Initial service targets (proposed)

Metric	Initial planning target
Website performance	Core pages within approximately 2.5 seconds at p75 under agreed test conditions
Backend latency	Common non-search API requests target under 500 ms at p95
Availability	Target 99.9% monthly uptime after operational maturity
Backup recovery point	Initially target no more than 15 minutes of lost database changes, subject to infrastructure
Recovery time	Initially target restoration within 4 hours for a major outage
Monitoring	Alert on payment failures, booking errors, supplier outages, and unusual traffic

These are proposed engineering goals, not guarantees. Validate them through load tests and confirm that the chosen infrastructure can support the recovery targets. Supplier API latency varies by provider and geography.

17. Testing and Acceptance Criteria

Test category	What must be verified
Unit tests	Price calculations, commissions, permissions, validation, state transitions
API integration	Database transactions, adapters, payment events, notification delivery
End-to-end	Customer journey from search through booking confirmation
Security	Access-control bypasses, injection, authentication weaknesses, exposed secrets, dependency risks
Payments	Success, decline, timeout, duplicate callback, refund, reconciliation
Suppliers	Price changes, expired offers, unavailable inventory, retries, duplicate requests
Multi-tenant	Agencies/suppliers cannot access unauthorized records
Performance	Concurrent searches, booking requests, queue processing, database load
Mobile	Android/iOS devices, connectivity loss, notifications, app lifecycle
Desktop	Windows/macOS installation, updates, secure storage, offline document access
Recovery	Database restoration, failed deployment, queue recovery, supplier outages
Acceptance	Agency staff, admins, and customers validate agreed workflows

Definition of done

A module is complete only when its interface, backend behavior, database persistence, permissions, validation, failure states, tests, and documentation are implemented. An attractive screen with nonfunctional buttons or simulated payments is not a completed feature.

Use demo supplier inventory and payment sandboxes until production access is approved. Clearly label test bookings and never issue fake live tickets or confirmations.

18. Phased Development Roadmap

Phase	Indicative timing	Deliverables
1. Discovery and foundation	Weeks 1–3	Business model, target markets, supplier evaluation, architecture, schema, repository, environments
2. Core marketplace	Weeks 4–8	Authentication, customer screens, search interfaces, agency/supplier onboarding, admin, core APIs, mock adapters
3. Booking engine and first integrations	Weeks 9–14	Offer normalization, revalidation, booking lifecycle, finance, payment sandbox, first provider adapter
4. Expand travel services	Weeks 15–20	Additional approved flight/hotel/car/tour/transfer integrations, packages, cancellations, refunds, reports
5. Mobile and desktop	Weeks 21–26	Android/iOS apps, Windows/macOS packages, notifications, documents, accessibility, device testing
6. Pilot and production launch	Weeks 27–30+	Agency onboarding, production approvals, security/performance assessment, controlled live tests, gradual launch

The schedule is an estimate for planning, not a guaranteed delivery date. Supplier onboarding, certification, commercial agreements, team size, and app-store approvals can extend it. Build the foundation and mock integrations while supplier applications are in progress.

Recommended launch sequence

- Build the architecture and all major service boundaries early.
- Validate the agency marketplace and supplier commercial model.
- Deliver a complete vertical booking flow using a mock adapter and payment sandbox.
- Replace the mock adapter with the first authorized provider integration.
- Expand to other service categories as access and certification are secured.
- Pilot with selected agencies, then scale based on measured reliability and demand.

19. Recommended Team

Role	Responsibility
Product manager / business lead	Requirements, business model, scope, priorities, acceptance
UI/UX designer	Website, mobile, desktop, admin, design system
Backend engineers	APIs, database, booking orchestration, finance, integrations
Web frontend engineer	Marketplace and management portals
Mobile engineer	Android and iOS application
Desktop engineer	Windows/macOS packaging, secure local behavior
QA engineer	Automated tests, integration testing, acceptance testing
DevOps/security engineer	Deployment, monitoring, secrets, recovery, security controls
Travel integration specialist	Supplier onboarding, API mapping, certification, operations

Some roles can be combined in a smaller team. Do not omit independent validation of supplier integrations, payment workflows, security, and booking recovery.

20. Risks and Mitigations

Risk	Mitigation
Supplier access delays or rejection	Apply early; build mock adapters and alternative provider options; never promise unavailable inventory.
Payment service unavailable in target market	Verify eligibility, settlement, and currencies before choosing the provider.
Payment and booking states diverge	Use idempotency, reconciliation, durable event handling, status checks, and recovery workflows.
Low agency adoption	Pilot with agencies, clear value proposition, transparent fees, and reliable support.
Refunds, disputes, currency exposure	Maintain reserves where appropriate, transparent terms, reconciliation, and qualified financial advice.
Different rules across six travel categories	Use service-specific workflows and policies behind shared booking abstractions.
Compliance gaps	Review target-market licensing, tax, consumer protection, privacy, and payment obligations before launch.
Security breach or data loss	Least privilege, encryption, monitoring, backups, restore tests, incident response.
Unrealistic delivery schedule	Prioritize vertical booking flow; phase service rollout; make external approvals explicit dependencies.

21. Project Kickoff Checklist

- **Approve business model:** Decide who collects customer payments, who issues tickets, and how agencies earn commissions.
- **Choose initial market:** Confirm launch countries, languages, currencies, and initial agency partners.

- **Confirm supplier access:** Identify candidate providers and request commercial and sandbox API access.
- **Build repository and database:** Create monorepo, authentication, tenant isolation, migrations, and CI pipeline.
- **Create UI and functional shell:** Build responsive customer screens, agency portal, admin portal, and shared navigation.
- **Implement first vertical booking flow:** Complete search, revalidation, booking, payment verification, confirmation, and error recovery for one authorized provider.

22. Final Delivery and Acceptance Checklist

- Complete source code and repository structure.
- Customer website and administrative dashboard.
- Agency and supplier portals.
- Android and iOS applications.
- Windows and macOS desktop applications, if included in the agreed scope.
- Database schema, migrations, seed data for development, and documented data relationships.
- API documentation and integration instructions.
- Supplier/payment integration capability register and configuration guides.
- Environment template without exposed secrets.
- Automated tests and test report.
- Deployment scripts, staging environment, and production instructions.
- Backup, monitoring, security, and recovery documentation.
- Customer, agency, supplier, and administrator user guides.
- Mobile/desktop build and release instructions.
- Clear list of features that require external accounts, credentials, subscriptions, provider approval, or commercial agreements.

Final acceptance rule: Demonstrate the complete agreed workflows, including search, price revalidation, booking, payment verification, supplier confirmation, cancellations/refunds, agency operations, reporting, and deployment. Any unavailable integration or demo-only feature must be explicitly documented.

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End of implementation plan.